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The Editor-in-Chief
IEEE Access

Dear Editor,

We submit our manuscript, “*Excitation-Coherent Telemetry Analysis for Predictive OBD-II Connector Wear Detection: A Physics-Informed Study on Real In-Vehicle Network Data*,” for consideration as a regular article in *IEEE Access*.

The SAE J1962 diagnostic connector now serves as the permanent mount and power source of telematics, insurance and fleet devices, although it was designed for occasional workshop use. Vibration-induced fretting of its contacts causes intermittent data loss and device brown-outs that are routinely blamed on the device or the vehicle. To our knowledge, this is the first study of whether the wear of the diagnostic connector itself can be detected, attributed and predicted from the telemetry that the attached device already records, without added hardware.

The main contributions are:

- a physics-informed model linking contact wear to the observable behaviour of three classes of diagnostic tool, with four realistic confounders;
- ECTA, a detection and attribution method built around a test of whether data losses co-vary with the vehicle’s own measured engine and road excitation;
- a first-principles observability law that predicts, without any classifier, which wear stages each class of tool can detect (Spearman 0.93 with the measured performance);
- an evaluation on five public real-vehicle datasets (nine vehicles in detail and a 381-vehicle fleet) with strictly vehicle-disjoint splits and eleven baseline and ablation models. On unseen vehicles a full-bus tool detects wear with AUROC 0.880 (moderate 0.891, severe 0.995) and confirms moderate wear within two minutes of driving, whereas deep sequence baselines reach 0.646.

We have taken care to report the limits of the approach as clearly as its successes: the connector faults are modelled on top of real telemetry (no public dataset of naturally worn J1962 connectors exists), the unmeasured constants of the model are stress-tested by deliberate mis-specification and by a structurally different fault process (on which a full-bus tool still reaches AUROC 0.856), and the negative findings (incipient wear, moderate wear for low-rate tools, and months-ahead remaining-life estimation) are reported in full. A Supplementary Material document reports the complete results, paired significance tests against every baseline, hyperparameter and window-length sensitivity, and false-alarm rates per hour of real driving. All code is available, and every reported number can be reproduced from it.

The work spans intelligent transportation, vehicular electronics, fault diagnosis and prognostics and health management, which we believe fits the broad, multidisciplinary scope of *IEEE Access*.

We confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have approved the manuscript and agree with its submission. The method

is related to Indian Patent Application 202641074361, filed by the authors, which is disclosed in the manuscript. The authors declare no other competing interests.

Sincerely,

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